

Approach	External trajectory/context	Role in training	Stronger teacher or expert signal
Expert/success-prefix RL	Expert, oracle, or successful trace	Provides a favorable starting state or a target to imitate	Usually required
Conventional weak-to-strong learning	Weak-model prediction	Serves as a pseudo-label or supervisory target	No stronger teacher, but the weak signal directly supervises the student
Hard-data construction	Newly synthesized or curated difficult problems	Expands task difficulty at the data level	Often requires strong generators, filtering, or substantial curation
DenoiseRL	Verifier-rejected trace from a weak model	Acts only as structured noise; the policy must reject/repair it and is updated only on its own continuation	Not required